

Optimized Project Task Scheduling System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Project Scheduling System is designed to efficiently manage and optimize the execution of tasks in a project while satisfying the dependencies between activities. Each task may depend on one or more other tasks, which must be completed before it can begin. The system uses data structures and optimization techniques to determine a valid and efficient execution order for all tasks. It generates an optimized project schedule with the objective of minimizing the overall completion time. The system also identifies critical tasks and the critical path, which are important for completing the project on time. The execution order of tasks is displayed clearly to help users understand the project workflow. This project demonstrates the practical application of concepts such as graphs, topological sorting, dependency management, and critical path analysis in project management. The system can help improve task planning, scheduling efficiency, and overall project management.

Signature of the Guide